FUP - Oral Exam

neslusam and question providers

Contents

⚠️ **TRIGGER WARNING - M*NAD** ⚠️ - The text below contains repeated, explicit occurences of the "M-Word".If you are currently recovering from trying to understand what a m*nad actually is, further review of this material is not recommended..

Introduction

[V] - Tomáš Votroubek , [H] - Rostislav Horčík , [Z] - Matěj Zorek **Your examiner is random, your previous lecturer doesn't make a difference.**

There might be a newer version of this document, check out this github repository

Functional Programming

[Z] What is pure/Higher-order function + examples of each possible combination?

Solution:
Pure function is like a mathematical function. Pure function is a **deterministic function without any side effects** (e.g., no modifying global variables, no writing to files)

Higher-order function: function taking **other functions as arguments** or **returning a function** or both is called a higher-order function.

order \ purity	Pure	Impure
First-Order	`circle_area r = pi * r * r`	`print x / getLine`
Higher-order	`map / filter / foldl / apply`	`mapM / modify (State monad)`

[Z] List types of recursions based on branching factor and provide examples. Classify head/tail recursion and discuss theirs branching factor.

Solution:
Recursion is split into **linear/tree recursion**.

Linear recursion makes **one** recursive call per execution step. Example: **Factorial**

Tree (Non-linear) recursion makes **two or more** recursive calls per execution step. Example: **Fibonacci**

In **head recursion**, the recursive call is made **before** the rest of the function's operations are completed. Head recursion is **usually non-linear**, because linear recursion is faster with tail recursion.

Example:

```
factorialHead 0 = 1
factorialHead n = n * factorialHead (n - 1)
```

In **tail recursion** the recursive call is the **very last** operation executed by the function. It often uses an **accumulator**. Tail recursion is always **linear**.

Example:

```
factorialTail 0 acc = acc
factorialTail n acc = factorialTail (n - 1) (n * acc)
```

Racket

[V] How to implement infinite stream of nats/ones

1. *using stream functions?*
2. *without using stream functions?*

Solution:
1. Example - stream of natural numbers using stream-cons

```
(define (stream_nats n)
(stream-cons n (stream_nats (+ n 1))))
```

2.

```
(define (nats_lambda n)
  (cons n (lambda () (nats_lambda (+ n 1)))))

;;for extra points use syntantic sugar thunk ;)

(define (nats_thunk n)
  (cons n (thunk (nats_thunk (+ n 1)))))
```

[V] Describe how does foldl/foldr work, which parametres are used and what is the difference between foldl and foldr.

Solution:
Foldl/r collapses a list into a single value, combining its elements with an accumulator from left to right (foldl) / from right to left (foldr).

Example:

```
(foldl append '() '((1 2) (3 4) (5 6)))
;; Output: '(5 6 3 4 1 2)

(foldr append '() '((1 2) (3 4) (5 6)))
;; Output: '(1 2 3 4 5 6)
```

λ-Calculus

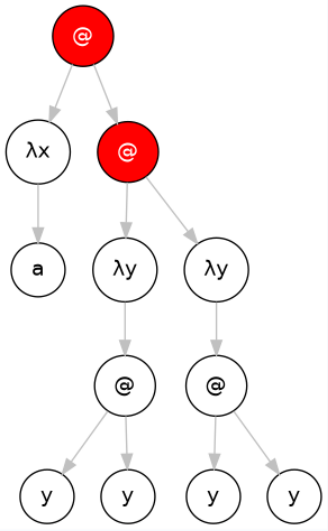
[Z] List and describe terms in syntax of λ-calculus.

Solution:
The syntax of lambda calculus has only three types of terms: a **variable** (denoted by lowercase letters x,y,z…), the **abstraction** of a variable from a term defining a function (denoted as λx.t), and the **application** of a term to a term (e.g. xy , (λx.y)y , …).

[H/Z] What is a redex?

Solution:
A **redex** (reducible expression) is any term that **matches abstraction to an argument**.It is similar to having a function (abstraction) that already got the argument and can be evaluated. It is really easy to find in an abstract syntax tree. Redex is **every application , where the left argument is an abstraction (lambda symbol)**.

Example: This AST has all the redexes marked red.



[H] What is a reduction? List and describe different types of reductions and their usage.

Solution:
Reduction is the step-by-step process of **evaluating** or simplifying an expression **until it cannot be simplified any further**.

The two essential transformation rules in lambda calculus are:

1. **α-conversion** - The process of **renaming a bound variable** (and all its occurrences inside the function body).Used to **prevent naming conflicts**

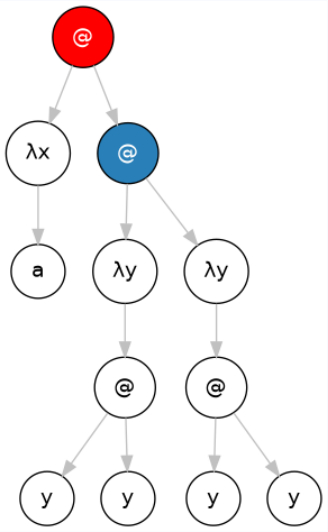
2.β-reduction - The actual computation step. It applies a function to an argument by **substituting the argument for all free occurrences of the bound variable inside the function body**.

There are **two** different strategies to evaluation:

1. Left outermost (Lazy evaluation) - furthest to the left and closest to the outside of the expression (**not contained inside any other redex**).

2. Left innermost (Eager evaluation) - furthest to the left but nested the deepest inside the expression (**it contains no other redexes inside itself**).

Example: **Red - left outermost , blue - left innermost**

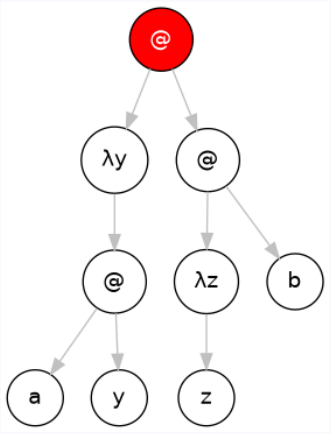


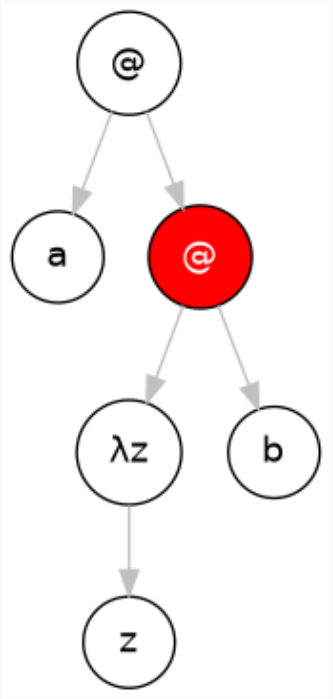
[V] What is a normal form and how to transform a λ-expression to its normal form?

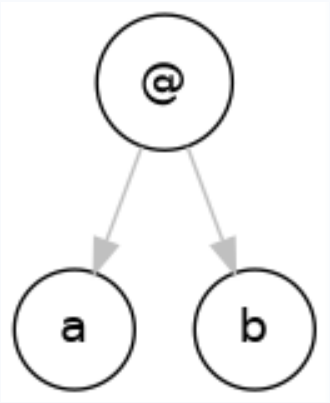
Solution:

Expression is in a **normal form** if it cannot be simplified any further. You can transform expression into a normal form using **repeated left outermost β-reductions** until there aren't left any.

Example: transformation of (λy. ay)((λ z. z)b)







Useful tips:

Be aware of name collision! Perform α-conversion before reducing to keep variable names unique.

Some expression do not converge and do not have a normal form! Example: (λ x.xx)(λx.xx)

[Z] What is the difference between bound and free variable? Which operation is done on them?

Solution:

Bound variable is inside of an abstraction, free is not.

Possible operations(probably what the teacher meant):

Bound variable - α-conversion - you can rename lambda and each instance in the body

Free variable - β-reduction - argument of a redex destroys the bound variables, only keeping the free ones.

[H] What is Church-Rosser theorem?

Solution:
TODO brandsi1

[H] how to represent True/False in λ-Calculus?

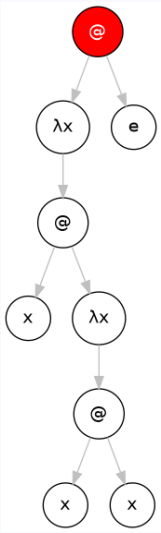
> **Solution:**
> **T = λxλy.x**
>
> **F = λxλy.y**
>
> Boolean is interpreted as a **decision between 2 options** and you either picks the first one (T) or the second one (F)

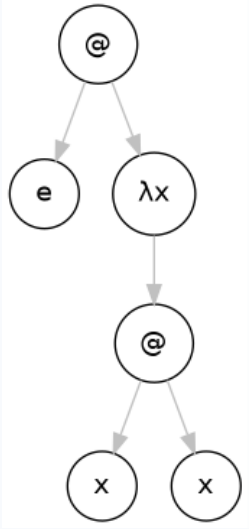
[H] How to represent number in λ-Calculus? How to represent addition of 2 numbers?

> **Solution:**
> **Number n is represented as applying function `f` to starting `x` n-times.**
>
> 0 = λ f. λ x. x 1 = λ f. λ x. f(x) 2 = λ f. λ x. f(f(x))
>
> addition is defined as:
>
> **ADD = λ m. λ n. λ f. λ x. m f (n f (x))**
>
> Example: ADD 1 2 = λ f. λ x. 1 f (2 f (x)) - **Note! We already defined what 1 and 2 means.**
>
> Argument of 1 f is (2 f (x)) = f (f (x))
>
> We are left with λ f. λ x. 1 f (f (f (x))), which is transformed back to λ f. λ x. f (f (f (x))) = **3**

[H] reduce this expression (λx.x(λx.xx))e to its normal form.

> **Solution:**
> This reduction because is tricky, because there are **two abstractions with the same name**, i would generally recommend to rename one to prevent mistakes from happening.

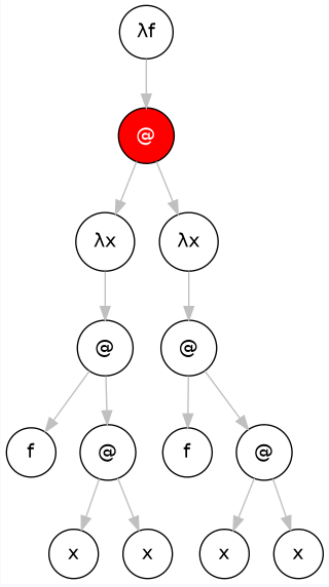


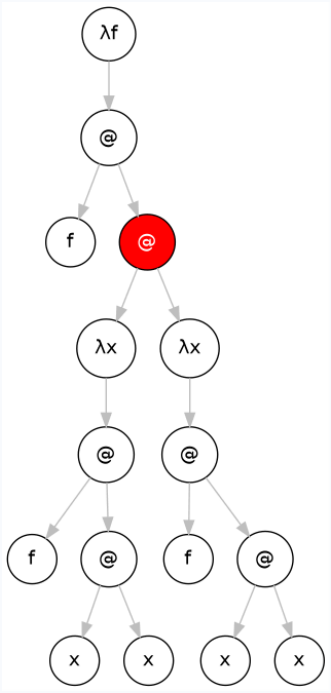


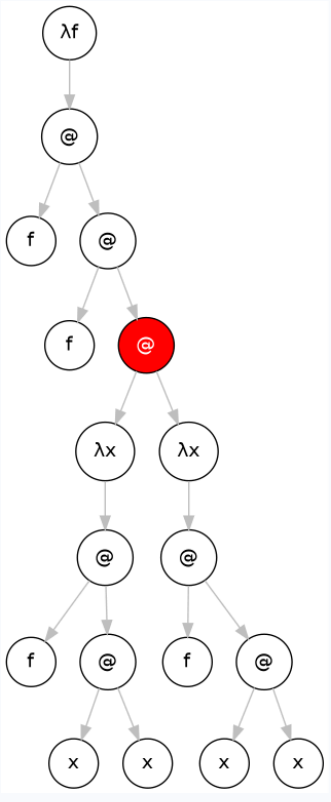
[V] Find all free/bound vars , redexes in (λ f. (λ x. f (x x)) (λ x. f (x x))). Find normal form.

Solution:

All variables are bound and there is only one redex (marked red). After a few steps, it is obvious that the term **grows infinitely and thus doesn't have a normal form**.







Haskell

[V] Describe signature of function map.

Solution:

```haskell
map :: (a -> b) -> [a] -> [b]
```

[V] Describe signature of function foldl/r.

Solution:

```haskell
foldr :: Foldable t => (a -> b -> b) -> b -> t a -> b
foldl :: Foldable t => (b -> a -> b) -> b -> t a -> b
```

[H] What is a typeclass?

Solution:
Typeclass is a sort of interface that defines some behavior. Typeclasses enable functions to work on different types while supporting certain operations.

For a custom data type , you can usually derive some typeclasses: `Eq, Ord, Enum, Bounded, Show, Read`. If this is not the case, you need to implement it yourself ;).

[H] How to implement polymorphic typeclass , which adds 1 to an Int and prints out length for String?

Solution:

You need to create a polymorphic typeclass with a single method that **returns IO monad**.

```haskell
class Magnesium a where
  wheatBran :: a -> IO ()

instance Magnesium Int where
  wheatBran n = do
    let next = n + 1
    print next

instance Magnesium String where
  wheatBran str = do
    let len = length str
    print len
```

[V] What is Functor typeclass? Which functions define it?

Solution:
Functor is any data type which can be mapped over. It enables applying pure functions without changing the structure.

Core function: `fmap :: (a -> b) -> f a -> f b` (or the `<$>` operator)

Example:

```
addOne = fmap (+1)

addOne (Map.fromList [('a',1), ('b',2)])
--Output: Map.fromList [('a',2), ('b',3)]
```

[V] What is a Applicative typeclass? Which functions define it?

Solution:
Applicative typeclass **extends Functor** typeclass. With a Functor, you use `<$>` to map a normal function over a wrapped value. With a **Applicative** typeclass, you can **apply a function that is already wrapped to a value that is also wrapped** using `<*>`, called **apply** or **ap**.

Another function of applicative typeclass is to **wrap in the simplest way possible** into a certain data type, using the function **pure**.

Core functions:

```
pure :: a -> f a

(<*>) :: f (a -> b) -> f a -> f b
```

Examples:

```
-- fmap fails here, because the function is wrapped
Just (+10) <$> Just 5  -- Result: Error

Just (+10) <*> Just 5  -- Result: Just 15
Nothing <*> Just 5  -- Result: Nothing

pure 5 :: Maybe Int   -- Output: Just 5
pure 5 :: [Int]   -- Output: [5]
```

[V] What is a Monad? Which functions define it?

Solution:
A Monad **extends Applicative** typeclass. Monads allow to **chain dependent operations**, meaning the output of **step A directly determines what step B will look like**. In contrast, Applicative executes a series of independent operations that are ignorant to each other until they are combined at the very end.

Core Functions:

1. **bind** (written as `>>=`): It feeds a boxed value to a function that returns a new box, without double-boxing.In the `do` notation , it is written with a arrow `<-`

```
(>>=) :: m a -> (a -> m b) -> m b
```

Example:

```
Just 10 >>= (\x -> Just (x + 1))  --Output: Just 11 (Unpacks 10, adds 1, repacks)
Nothing >>= (\x -> Just (x + 1))  --Output: Nothing

Just 10 <*> (\x -> Just (x + 1))  --Output: Error (Applicative is not enough)
```

2. **>> operator**: >> chains two monadic actions together by running the first, discarding its value, and then running the second. This seems useless at first, but under the hood, every time you write a line in a do block without using `<-`, Haskell translates it to >> under the hood.

```
(>>) :: m a -> m b -> m b
```

Example:

```
--This is equvalent:
main = do
  putStrLn "Hello"
  putStrLn "World"

--to this:
main = putStrLn "Hello" >> putStrLn "World"
```

3. **return**: the **same function as `pure`** in Applicative typeclass

```
   return :: a -> m a
```

Example from 1. rewritten with do notation.

```
successExample :: Maybe Int
successExample = do
  x <- Just 10
  return (x + 1)
```

[H] What is a State Monad, what is it good for? How to implement bind for State Monad?

Solution:
State Monads simulate mutable state in pure functional programming. State is just a function that takes an initial state s, and returns a tuple containing the result a and a new state s.

```
State s  a = s -> (a, s)
```

Even though a student has only been asked about bind, I highly recommend looking into State.hs provided in the official page. In my opinion, the question about bind open possibilities to be asked about different functions.

```haskell
newtype State s a = S { runState :: s -> (a, s) }

instance Monad (State s) where
  -- (>>=) :: State s a -> (a -> State s b) -> State s b
  stx >>= f = S (\s -> let (x, s') = runState stx s
                       in runState (f x) s')
```

You are all done!